INCH-POUND
MIL-DTL-16377/76A(SH)
26 May 2023
SUPERSEDING
MIL-DTL-16377/76(SH)
27 September 2004

DETAIL SPECIFICATION SHEET

FIXTURES, LIGHTING; FLUORESCENT AND SOLID-STATE EQUIVALENT, EMERGENCY LIGHTING, 60 AND 80 WATTS, 120 VOLTS, 60 HERTZ, DRIPPROOF, SYMBOLS 348EM, 349EM, 348EMSSL, AND 349EMSSL

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16377.

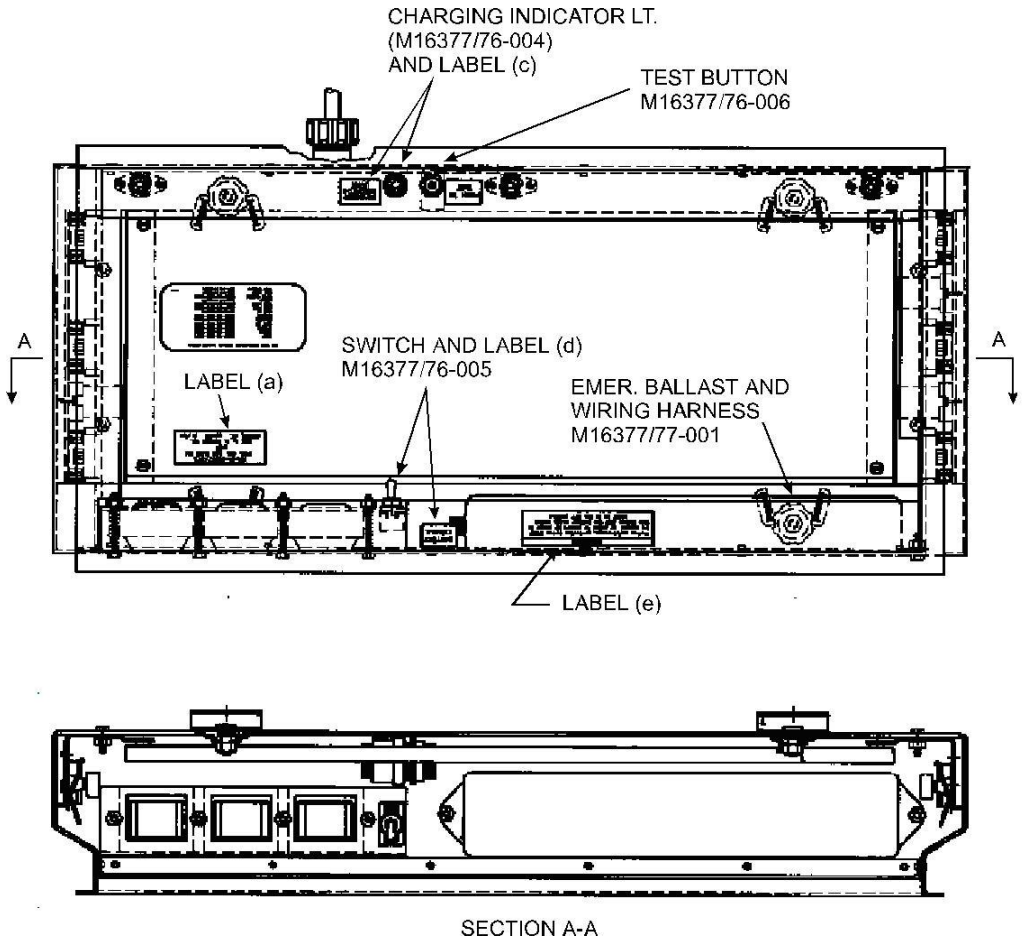
Unless otherwise noted, all dimensions are in inches. For tolerances other than those specified herein, see 3.5.2 of MIL-DTL-16377.

Figures 1 through 4 provide information for the light fixtures covered by this specification sheet. See the requirements sections for specific references to these figures.

AMSC N/A FSC 6210

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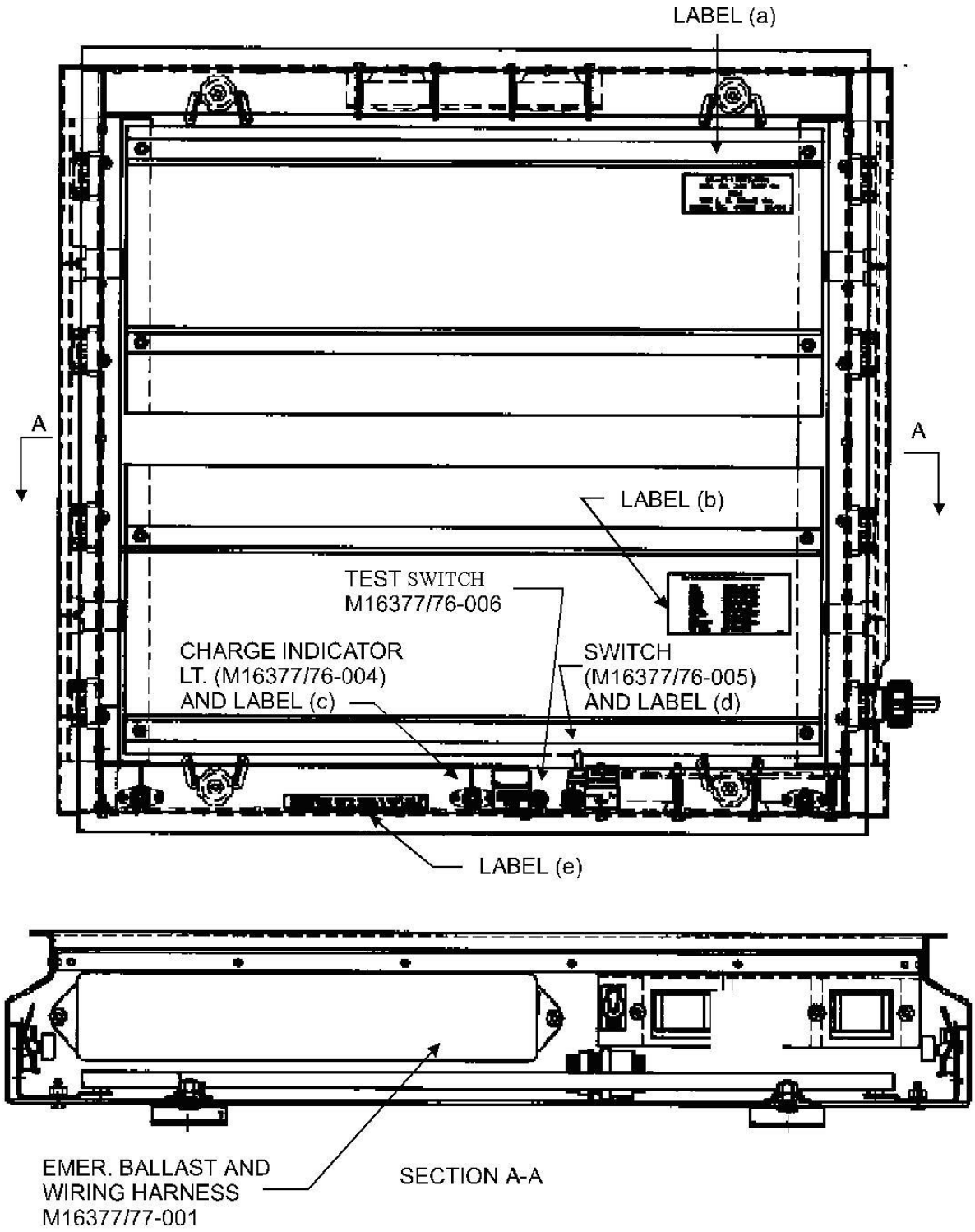
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NOTE: For dimensional data, see figures 2 and 3 of MIL-DTL-16377/57.

FIGURE 1. Typical view of symbol 348EM.

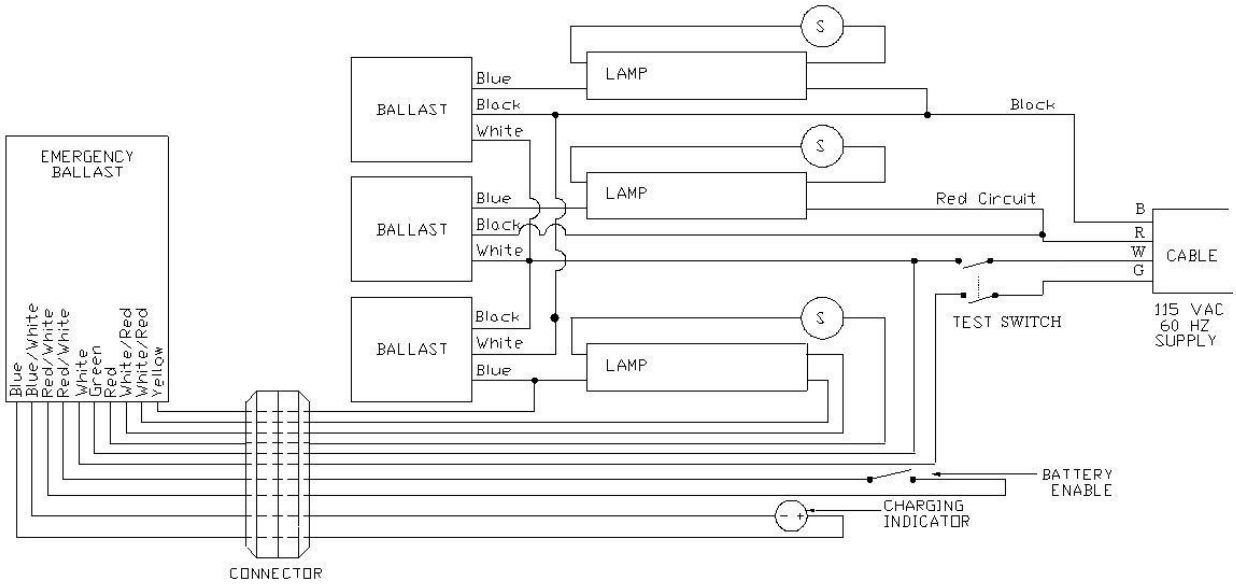
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NOTE: For dimensional data, see figures 2 and 3 of MIL-DTL-16377/57.

FIGURE 2. Typical view of symbol 349EM.

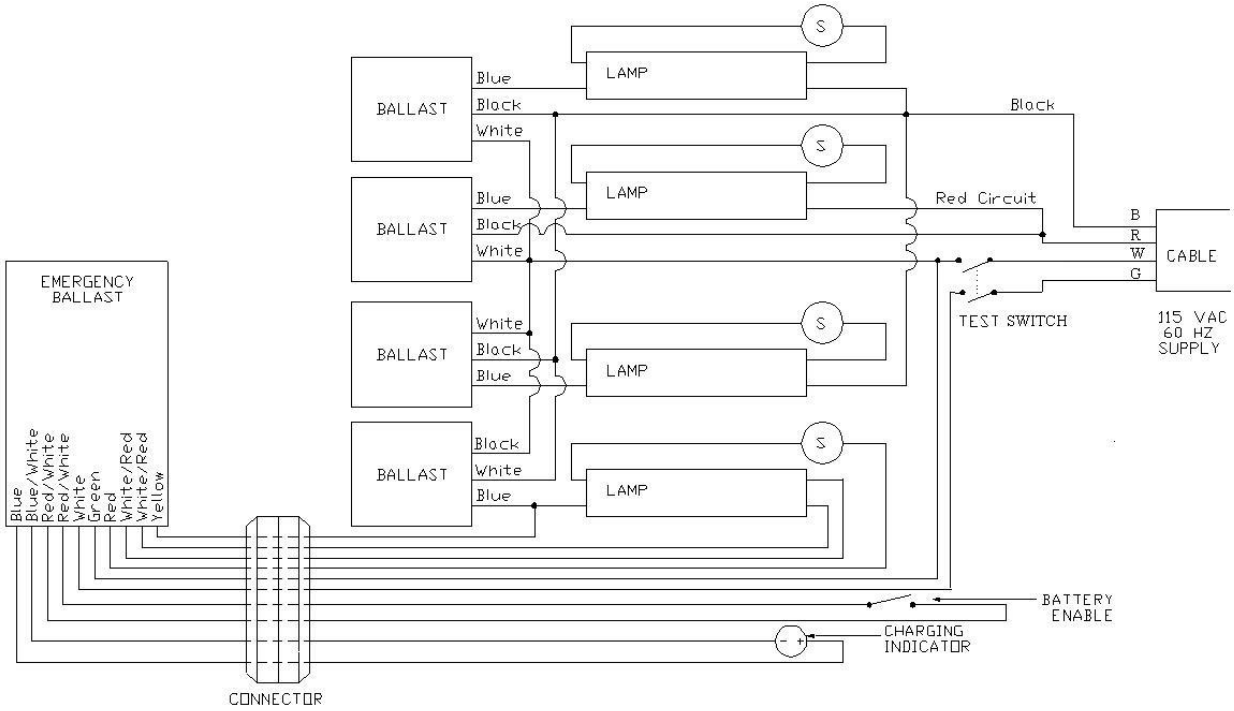
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NOTES:

1. Ballast black and white wiring not color coded with input cable black and white, emergency lamp only.
2. Emergency ballast white and green wiring not color coded with input cable white and green.

FIGURE 3. Wiring diagram for symbol 348EM.



NOTES:

1. Ballast black and white wiring not color coded with input cable black and white, emergency lamp only.
2. Emergency ballast white and green wiring not color coded with input cable white and green.

FIGURE 4. Wiring diagram for symbol 349EM.

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FLUORESCENT REQUIREMENTS

CLASSIFICATION

Type: I
Class: 2

SYMBOL NUMBERS

348EM: Complete lighting fixture assembly, flush mounting with three 20-watt cool white fluorescent lamps, clear prismatic window, and fluorescent emergency ballast.

349EM: Complete lighting fixture assembly, flush mounting with four 20-watt cool white fluorescent lamps, clear prismatic window, and fluorescent emergency ballast.

DESIGN

Electrical rating:	60 and 80 watts, 120 volts, 60 hertz.
Dimensions and configuration:	See figures 1, 2, 3, and 4 of MIL-DTL-16377/57.
Photometric data:	See figures 11 and 12 of MIL-DTL-16377/57.
Lampholder spacing:	See figure 21 of MIL-DTL-16377.
Wiring diagram:	See figure 3 for symbol 348EM or figure 4 for symbol 349EM.
Mounting dimensions:	
Symbols 348EM and 349EM:	See figure 10 of MIL-DTL-16377/57.
Enclosure:	Dripproof, 15 degrees.
Cable:	10 feet of type LSFHOF-3 in accordance with MIL-DTL-24643/3.
Weight:	
Symbol 348EM:	21 pounds maximum
Symbol 349EM:	28 pounds maximum
Designation and marking:	The following labels shall be furnished in accordance with MIL-DTL-16377. Labels (a) and (b) shall be in accordance with MIL-DTL-16377/57. Labels (c), (d), and (e) shall be as shown on figures 1 and 2. (a) Type I labels. (b) One type IV label. One type IV label shall be applied as shown on figures 1 and 2. The information to be shown on the type IV label shall be: "USE THE FOLLOWING REPAIR PARTS ONLY". List all standard parts shown in table I by name and stock number only. (c) One type IV label. The information to be shown on the type IV label shall be: "CHARGING INDICATOR LIGHT". (d) One type IV label. The information to be shown on the type IV label shall be: "BATTERY ENABLE". (e) One type V label. The information to be shown on the type V label shall be: "DURING MAINTENANCE, PROTECTIVE DEVICE MUST BE OPENED OR PULLED TO SECURE AC POWER TO THIS FIXTURE AND THE BATTERY ENABLE SWITCH MUST BE IN THE OFF POSITION".
Standard parts:	The standard parts and their respective part numbers shown in table I shall be used in the construction of the fixtures specified herein.

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TABLE I. Standard parts.

Military Part Number	Description	Specification/Materials	Remarks
M16377/76-001	Lighting assembly	M16377/57-348	For use with symbol 348EM
M16377/76-002	Lighting assembly	M16377/57-349	For use with symbol 349EM
M16377/76-003	Mating harness	AMP part No. 206043-1, or equal	Mates with M16377/77-001
M16377/76-004	Indicator light	Commercial	LED-type
M16377/76-005	Switch (DPST)	MS16569	---
M16377/76-006	Test Switch	Commercial	Double pole, test switch with rubber boot
M16377/77-001	Emergency ballast with wiring harness	AMP part No. 206043, or equal	Mates with M16377/76-003
NOTE: Switch shall normally be left in the “Battery Enable” position.			

SOLID-STATE LIGHTING (SSL) REQUIREMENTS

CLASSIFICATION

Type: III
Class: 2

SYMBOL NUMBERS

348EMSSL: Complete lighting fixture assembly, standard mounting with LED light source, clear prismatic window assembly, and emergency battery backup.
349EMSSL: Complete lighting fixture assembly, standard mounting with LED light source, clear prismatic window assembly, and emergency battery backup.

DESIGN

Electrical rating:
348EMSSL: 30 watts maximum, 120 volts, 60 hertz.
349EMSSL: 40 watts maximum, 120 volts, 60 hertz.
Dimension and configurations: External dimensions shall not exceed those shown on figures 2 through 4 of MIL-DTL-16377/57 and figure 10 of MIL DTL-16377.
Photometric data: See figures 11 and 12 of MIL-DTL-16377/57 and 3.7.1 of MIL-DTL-16377.
Emergency mode lumen output: Luminaire shall provide a minimum of 240 lumens on a fully charged battery and maintain over 144 lumens over a 5-hour runtime.
Chromaticity: See 3.7.4 of MIL-DTL-16377.
Fail-safe circuit: See 3.17 of MIL-DTL-16377.
Wiring: Wiring shall maintain same control scheme as legacy fluorescent, including battery enable switch, battery charging indicator.
Mounting dimensions:
Symbols 348EMSSL and 349EMSSL: See figure 10 of MIL-DTL-16377/57.
Enclosure: Dripproof, 15 degrees. See 3.5.11 of MIL-DTL-16377.
Battery backup requirements: See 3.6.19 of MIL-DTL-16377.
Cable: 10 feet of low-smoke cable in accordance with MIL-DTL-24643/3; sufficient conductors to comply with control scheme and grounding requirements.

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Weight:
Symbol 348EMSSL: 21 pounds maximum.
Symbol 349EMSSL: 28 pounds maximum.
Designation and marking: The following labels shall be furnished in accordance with MIL-DTL-16377.
(a) Two type I.
(b) One type IV. The information to be shown on the type IV label shall be: "USE THE FOLLOWING REPAIR PARTS ONLY," followed by a list of available replacement parts and stock numbers; or if unavailable, manufacturer's part numbers.

EXAMINATIONS AND TESTS

Examinations and tests shall be as specified in MIL-DTL-16377 and as follows:

First article inspection:
Complete fixtures: See table II for fluorescent and table III for SSL.
Conformance inspection:
Comparison inspection: Same as the first article inspection.
Conformance inspection: See MIL-DTL-16377.

TABLE II. First article and comparison inspection for complete type I fixtures.

Tests	Requirement Paragraph of MIL-DTL-16377	Test Paragraph of MIL-DTL-16377	Remarks
Examination	3.12	4.5	---
Operation	3.12	4.8.1	---
Dielectric withstanding voltage	3.6.15	4.8.2	---
Insulation resistance	3.6.16	4.8.3	---
Enclosure effectiveness	3.5.11	4.8.14.4	Dripproof
Shock	3.5.12	4.8.7	---
Vibration	3.5.13	4.8.8	---
Enclosure effectiveness	3.5.11	4.8.14.4	Dripproof
Dielectric withstanding voltage	3.6.15	4.8.2	---
Insulation resistance	3.6.16	4.8.3	---
Noise	3.5.14	4.8.9	---
Continuity of grounding	3.6.13.4	4.8.12	---
Light output	3.7.1	4.8.6.1	Candlepower
Optical uniformity	3.7.4.1.1	4.8.4	---
Stress relief	3.8.5.3	4.8.5	---
Magnetic permeability	3.4.8	4.8.18	---

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TABLE III. First article and comparison inspection for complete type III fixtures.

Tests	Requirement Paragraph of MIL-DTL-16377	Test Paragraph of MIL-DTL-16377	Remarks
Examination	3.12	4.5	---
Operation	3.12	4.8.1	---
Power interface	3.6.18	4.8.21	---
Emergency light output	3.7.1.1	4.8.6.8	Lumen output
Emergency battery back-up performance: Operating time and discharge test	3.6.19.3	4.8.23	---
Electromagnetic interference	3.6.14	4.8.16	---
Dielectric withstanding voltage	3.6.15	4.8.2	1/
Insulation resistance	3.6.16	4.8.3	1/
Enclosure effectiveness	3.5.11	4.8.14	1/, 2/
Shock	3.5.12	4.8.7	1/
Vibration	3.5.13	4.8.8	1/
Enclosure effectiveness	3.5.11	4.8.14	1/, 2/
Dielectric withstanding voltage	3.6.15	4.8.2	1/
Insulation resistance	3.6.14	4.8.3	1/
Noise	3.5.14	4.8.9	1/
Continuity of grounding	3.6.13.4	4.8.12	---
Photometrics	3.7.1	4.8.6	Candlepower
Brightness	3.7.5	4.8.6.4	---
Chromaticity	3.7.4	4.8.6.5	---
Correlated color temperature	3.7.4.1.1	4.8.6.6	---
Lumen maintenance	3.7.11	4.8.20	---
Fail-safe circuit design	3.17	4.8.24	---
Optical uniformity	3.7.7.1	4.8.4	---
Stress relief	3.8.5.3	4.8.5	---
Magnetic permeability	3.4.8	4.8.18	---

NOTES:
1/ In accordance with 4.3.1 of MIL-DTL-16377, tests shall be performed on one luminaire in the order specified.
2/ Dripproof, 15 degrees.

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CHANGES FROM PREVIOUS ISSUE. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 6210-2023-019)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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